LAB EXPERIMENT RECORD

03-10-2026---Set 7: Customer Demographics Analysis

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Register Number	192325114
Date	03 OCTOBER 2026
Day	Saturday

Aim

To analyze customer demographic data using a bar chart, pie chart and table, and to build an interactive Tableau dashboard for exploring customer demographics.

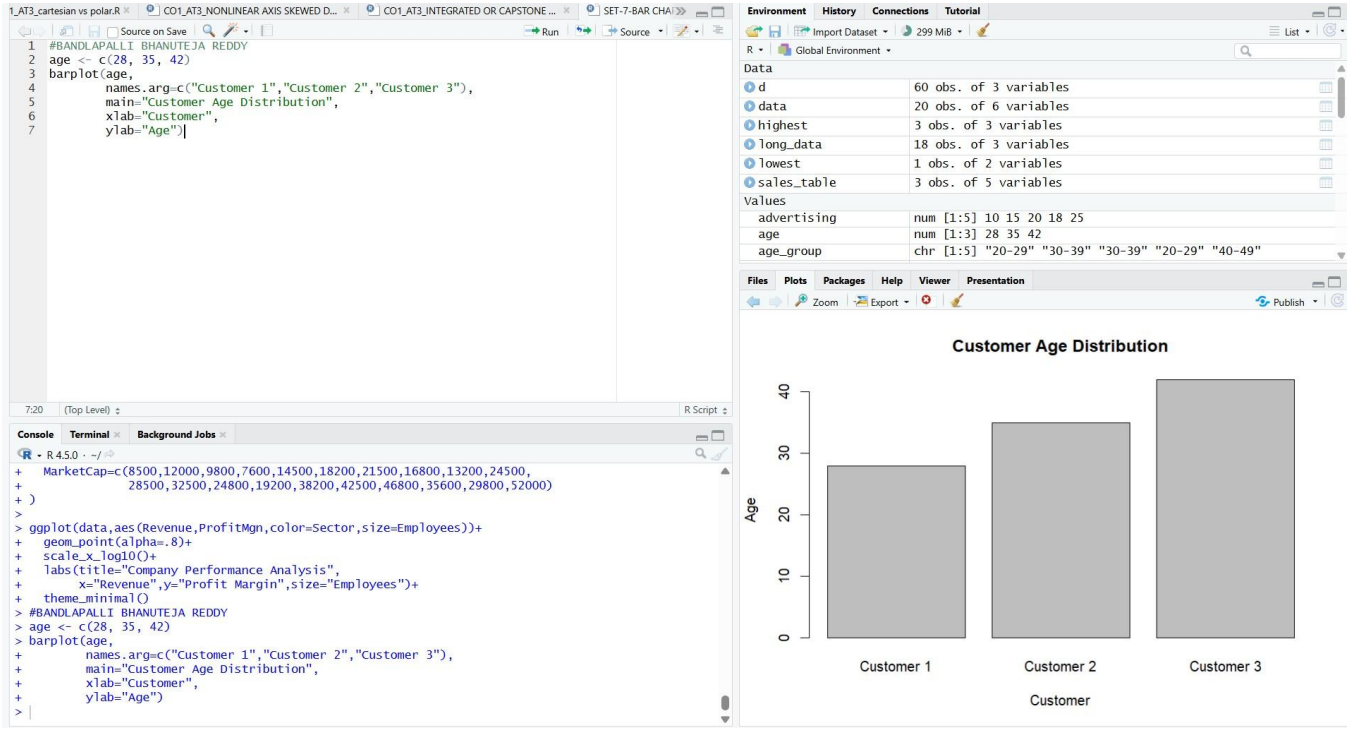
Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in $)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

Question 1: Bar Chart of Customer Ages

Question: Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

Aim: To visualize the distribution of customer ages using a bar chart with labelled axes and a title.

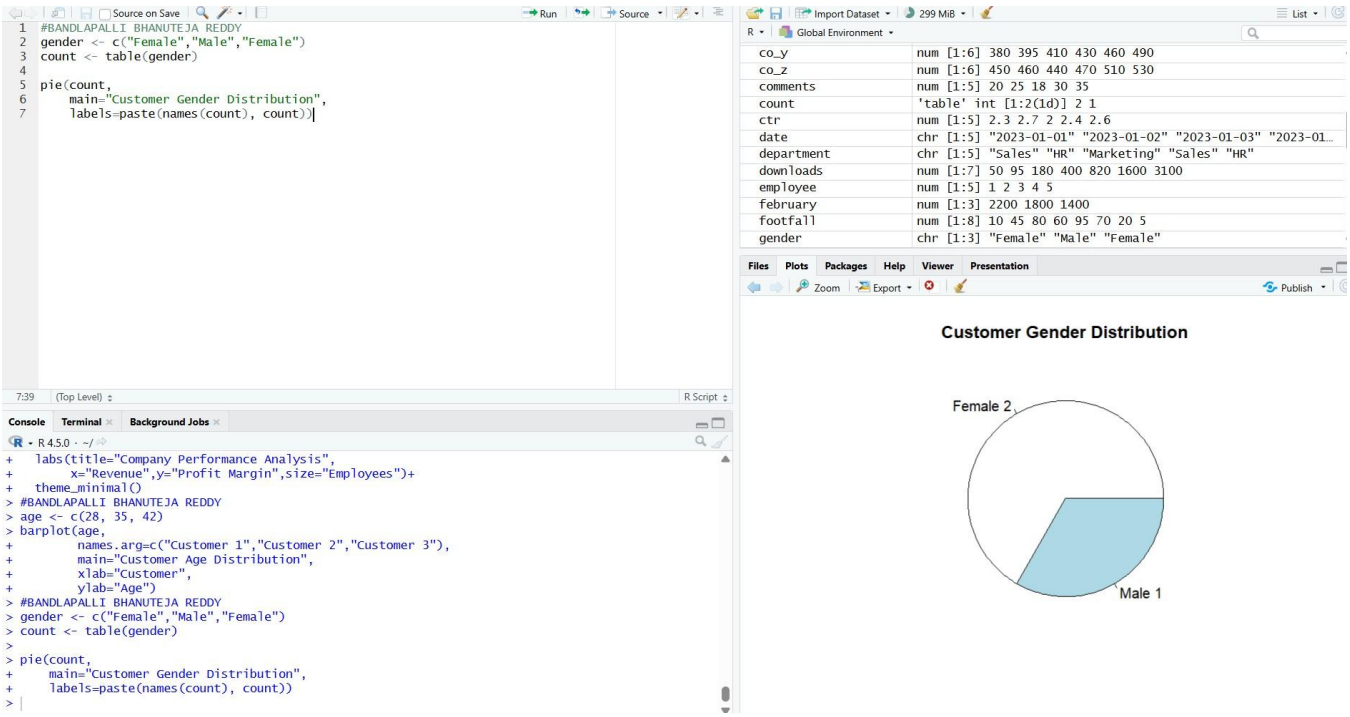


Result: The three customers are aged 28, 35 and 42, so the ages are spread evenly across a 14-year range with one customer at each age.

Question 2: Pie Chart of Customers by Gender

Question: Generate a pie chart to represent the distribution of customers by gender.

Aim: To represent the share of customers by gender using a labelled pie chart.



Result: Female customers form 66.7% (2 customers) and male customers form 33.3% (1 customer).

Question 3: Table of Demographic Information

Question: Build a table to show the demographic information for each customer. Label the table elements.

Aim: To present the age, gender and income of each customer in a clearly labelled table.

TEGRATED OR CAPSTONE ... | SET-7-BAR CHART CUSTOMER AGES.R | SET-7-PIE CHART GENDER DISTRIBUTI...

Source on Save | Run

```
1  #BANDLAPALLI BHANUTEJA REDDY
2  customer_id <- c(1,2,3)
3  age <- c(28,35,42)
4  gender <- c("Female","Male","Female")
5  income <- c(50000,60000,75000)
6
7  data <- data.frame(customer_id, age, gender, income)
8  print(data)
```

8:12 (Top Level)

Console | Terminal | Background Jobs

R 4.5.0 · ~/

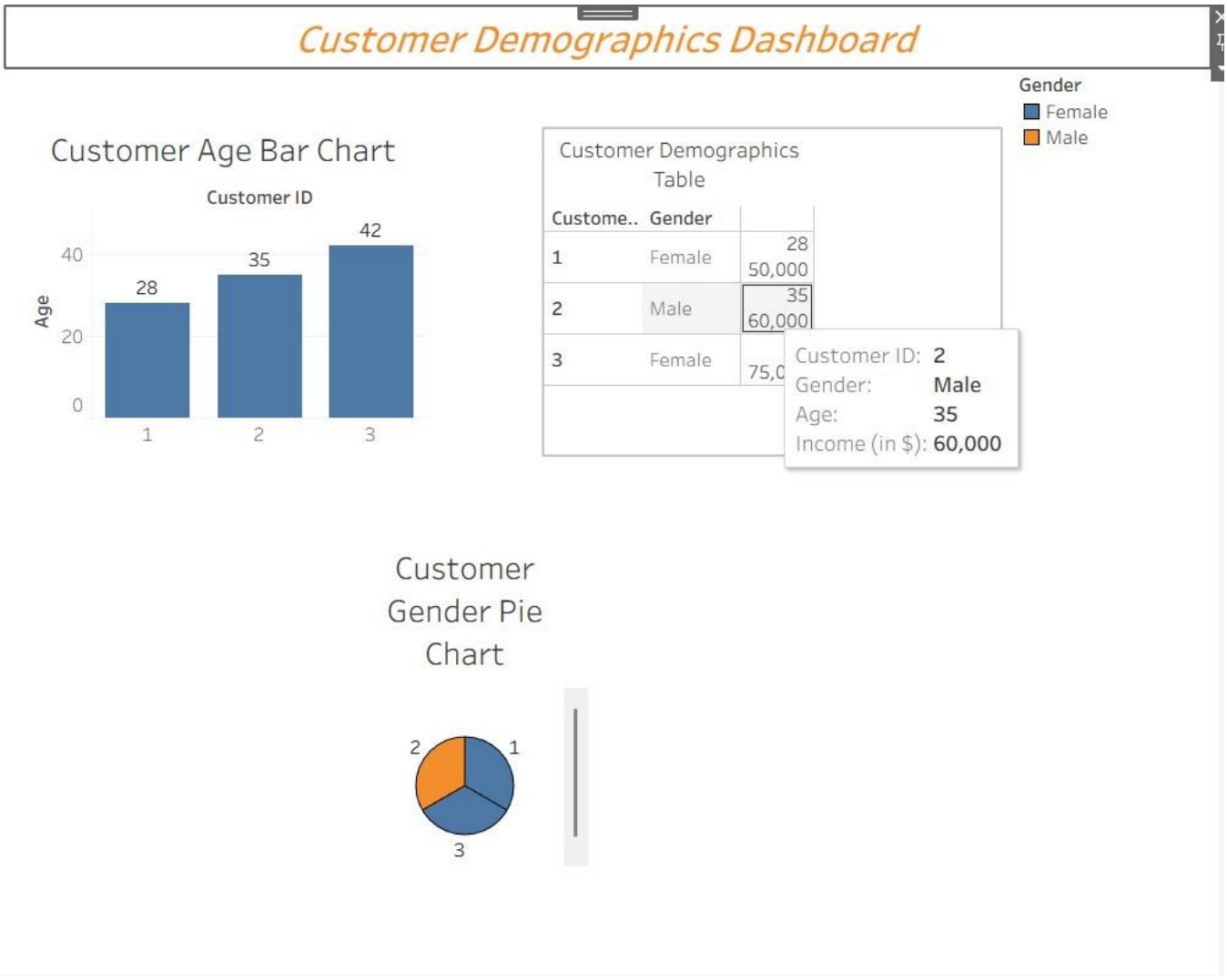
```
> print(data)
  customer_id age gender income
1           1  28 Female  50000
2           2  35   Male  60000
3           3  42 Female  75000
> #BANDLAPALLI BHANUTEJA REDDY
> customer_id <- c(1,2,3)
> age <- c(28,35,42)
> gender <- c("Female","Male","Female")
> income <- c(50000,60000,75000)
>
> data <- data.frame(customer_id, age, gender, income)
> print(data)
  customer_id age gender income
1           1  28 Female  50000
2           2  35   Male  60000
3           3  42 Female  75000
>
```

Result: The table lists all three customers with their details; Customer 3 has the highest income (75000) and Customer 1 the lowest (50000).

Question 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

Aim: To build an interactive Tableau dashboard that combines the bar chart, pie chart and table for exploring customer demographics.



Result: The dashboard places all three views together, and a gender filter updates every view so users can explore the demographics interactively.

Overall Result

Observation: The average customer age is 35 years and the average income is about 61,667, with income rising as age increases.